Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.15	20.12	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.01	-0.2%
power (W), mean	630	629	-0.1%
power (W), peak	652	662	+1.4%
energy (Wh)	212	211	-0.3%
energy, parked workers still on at idle power (Wh)	212	211	-0.3%
CPU used (cores), mean	0.914	0.886	-3.1%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.038	0.037	-3.1%
energy per core-hour (Wh)	689	710	+3.1%
node-hours per core-hour	6.56	6.77	+3.2%
pending pods, pod-minutes	0.533	0.267	-50.0%
pending pods, peak	2.00	1.00	-50.0%
HPA replicas, mean	8.92	9.06	+1.6%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	1.30	0.533	-59.0%
requests timed	1,035	1,145	+10.6%
time over the response line (% of samples)	0.870	0.000	-100.0%
response time (ms), mean	161	49.04	-69.5%
response time (ms), median	130	43.00	-67.0%
response time (ms), 95th percentile	324	70.30	-78.3%
response time (ms), 99th percentile	451	103	-77.1%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.1993	7.6345	+6.0%	[-1.5577, +2.3847]	EVEN, no significant difference
kwh_per_core_hour	0.7524	0.7957	+5.8%	[-0.1446, +0.2369]	EVEN, no significant difference
utilisation	0.0377	0.0366	-2.9%	[-0.0112, +0.0097]	EVEN, no significant difference
pending_pod_minutes_per_hour	1.5238	0.7805	-48.8%	[-4.5714, +2.3415]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	3.7619	1.5485	-58.8%	[-6.7436, +1.6083]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.358 0.391 0.42 0.444 0.466 0.485 0.502 0.517 0.53 0.541 0.55 0.557 0.562 0.566 0.571 0.575 0.579 0.583 0.585 0.586
- nodes decided per minute: 6 6 5 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	616	616	4.25	5.00	0.000	0.000	0.546	0.545

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	620	615	5.00	5.75	0.000	0.000	0.622	0.541
2	6.00	6.00	616	615	6.00	6.00	0.000	0.000	0.548	0.505
3	6.00	6.00	623	622	6.00	6.00	0.500	0.250	0.709	0.683
4	6.00	6.00	633	634	8.50	9.00	0.000	0.000	1.02	1.01
5	6.00	6.00	636	632	9.33	10.00	0.000	0.000	1.08	0.947
6	6.00	6.00	638	631	10.00	10.00	0.000	0.000	1.10	0.949
7	6.00	6.00	648	652	10.00	10.00	0.000	0.000	1.40	1.54
8	6.00	6.00	647	650	10.00	10.00	0.000	0.000	1.36	1.46
9	6.00	6.00	648	656	10.00	10.00	0.000	0.000	1.39	1.60
10	6.00	6.00	647	646	10.00	10.00	0.000	0.000	1.36	1.36
11	6.00	6.00	618	618	10.00	10.00	0.000	0.000	0.622	0.598
12	6.00	6.00	619	616	10.00	10.00	0.000	0.000	0.642	0.543
13	6.00	6.00	627	622	10.00	10.00	0.000	0.000	0.816	0.675
14	6.00	6.00	636	634	10.00	10.00	0.000	0.000	1.08	1.04
15	6.00	6.00	636	634	10.00	10.00	0.000	0.000	1.05	0.926
16	6.00	6.00	632	636	10.00	10.00	0.000	0.000	0.987	1.05
17	6.00	6.00	627	629	10.00	10.00	0.000	0.000	0.865	0.869
18	6.00	6.00	620	616	10.00	10.00	0.000	0.000	0.605	0.525
19	6.00	6.00	620	616	10.00	10.00	0.000	0.000	0.632	0.549

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.